The negative logarithms in your `Other_terms.tex`

$A \equiv (x-z)^2$, $B \equiv (x-w)^2$, $a \equiv \frac{k^+}{\vee}$. Blue = new, red = removed. Corrected file: `corrected/Other_terms.tex` (clean) and `corrected/Other_terms_marked.tex` (marked).

1 What changes

Fifteen logarithms, all in Group I Row III, have arguments that change sign in the transverse plane. Each becomes $\log|\cdot|$:

where	logarithm	number	negative where
part 1, line 2	$\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-z)^2-(x-w)^2}\right] \to \log\left|\cdot\right|$	1	$aB < A < B$
part 1, line 3; part 3a (6 lines)	$\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right] \to \log\left|\cdot\right|$	7	$A > B$
part 1, line 4	$\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right] \to \log\left|\cdot\right|$	1	$A < aB$
part 2, lines 3–8	$\log\left[\frac{(x-w)^2\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right] \to \log\left|\cdot\right|$	6	$aB < A < B$

- **Part 1, line 4.** Your new version $\log\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}$ is right for $(x-z)^2 > \frac{k^+}{\vee}(x-w)^2$. It is negative in the small region $(x-z)^2 < \frac{k^+}{\vee}(x-w)^2$ (near $x=z$). There the old sign was the right one, so the general form is the modulus.
- **The rest of the file.** Every other logarithm has a positive argument everywhere:
  - $\log(1-\frac{k^+}{\vee})$, $\log(1-\frac{\Lambda}{k^+})$, $\log\frac{(z-w)^2}{(x-w)^2}$;
  - all logarithms of $(1-\frac{k^+}{\vee})(x-z)^2+\frac{k^+}{\vee}(x-w)^2$ (Group I Row IV, Group II Row I, Group III);
  - the two logarithms in R^i, where $\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right) = (1-\frac{k^+}{\vee})\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee} > 0$.

  The arctan function $\mathcal{A}_0$ is real. Nothing else changes.

2 Why the modulus

Every one of these logarithms comes from a p^+ integral of Row III with the denominator $p^+(x-z)^2 - k^+(x-w)^2$. It vanishes at

$$p_0^+ = k^+\frac{(x-w)^2}{(x-z)^2}.$$

The argument is negative exactly when p_0^+ lies inside the range of that integral, as the table shows:

- part 1 line 2 and part 2: $(k^+, \vee)$;
- part 1 line 3 and part 3a: $(0, k^+)$;
- part 1 line 4: $(0, \vee)$.

In your calculation this denominator comes from the Fourier transforms (i3)–(i5), e.g. $\int \frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\frac{e^{ik\cdot X+ip\cdot Z}}{\xi k^2-p^2} = \frac{1}{\xi Z^2-X^2}$. These are principal values. With the principal value the p^+ integral is

$$\mathrm{PV}\int_{p_1}^{p_2}\frac{dp^+\,A}{p^+A-k^+B} = \log\left|\frac{p_2A-k^+B}{p_1A-k^+B}\right|,$$

which gives the moduli of the table. The $\log\frac{\vee}{k^+}$ and $\log\frac{k^+}{\Lambda}$ terms you split off are unchanged.

Check (`pv_check.py`). For each of the four logarithms, the numerical principal-value integral minus the split-off $\log \frac{\vee}{k^+}$ equals $\log|\cdot|$. This holds in all three regions $A < aB$, $aB < A < B$, $A > B$, to 4×10^{-12}. My earlier first-principles check of Row III (`Group_I_check.pdf`, G4) found the same: with $\log|\cdot|$ every line agrees with the principal-value integral of the line it comes from.

About the $i\pi$ of my previous note. An $i\pi$ term appears only if the momentum denominator $\xi k^2 - p^2$ is taken with $\pm i\varepsilon$ instead of the principal value. That is a choice in (i3)–(i5), not in Row III. With your principal-value transforms there is no $i\pi$, and the moduli are the complete fix. If you change (i3)–(i5) to $\pm i\varepsilon$, every term built from them changes, not only these logarithms.

Earlier checks. My numerical checks of these terms (divergences, `UV_divergences.pdf`) already used $\log|\cdot|$, so they hold for the corrected file.